

Algebra II with Trigonometry

Chapter 11.2

Mental Math (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4666322_alg-2trig-h-chp-11-2-note-docx.pdf](#)

Major Axis Length 1. What is the length of the major axis for the ellipse with the equation $\frac{x^2}{25} + \frac{y^2}{9} = 1$?

Minor Axis Length 2. For the ellipse $\frac{x^2}{16} + \frac{y^2}{49} = 1$, what is the length of the minor axis?

Ellipse Orientation 3. Does the ellipse given by the equation $\frac{x^2}{10} + \frac{y^2}{36} = 1$ have its major axis on the x-axis or the y-axis?

Focal Distance 4. If an ellipse has the equation $\frac{x^2}{25} + \frac{y^2}{16} = 1$, what is the distance from the center to a single focus (the value of c)?

Calculating Eccentricity 5. What is the eccentricity of the ellipse $\frac{x^2}{100} + \frac{y^2}{64} = 1$?

Eccentricity of Zero 6. If the eccentricity of an ellipse is exactly 0 ($a = b$), what specific geometric shape does the graph form?

Standard Form Conversion 7. If you rewrite the equation $x^2 + 4y^2 = 4$ into standard ellipse form, what will be the value of the larger denominator?

Finding the Minor Denominator 8. An ellipse has its center at the origin, vertices at $(\pm 5, 0)$, and foci at $(\pm 4, 0)$. What is the smaller denominator (b^2) in its standard equation?

Vertices to Axis Length 9. If the vertices of a vertical ellipse are located at $(0, 7)$ and $(0, -7)$, what is the total length of its major axis?

Distance Between Foci 10. For the ellipse $\frac{x^2}{169} + \frac{y^2}{144} = 1$, what is the total distance between its two foci?